

Database Management System

Semester-III (Batch-2024)

Multi- Tool AI Platform



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1. Introduction

The **Multitool AI Platform** is an innovative application that leverages Artificial Intelligence to provide users with powerful tools for content creation, automation, and management. It enables users to generate text, create images, and manage creative outputs seamlessly, along with social engagement features like liking content.

To ensure smooth functionality, security, and scalability, the platform relies on a **well-structured and normalized relational database** as its backbone. This database is designed to efficiently manage:

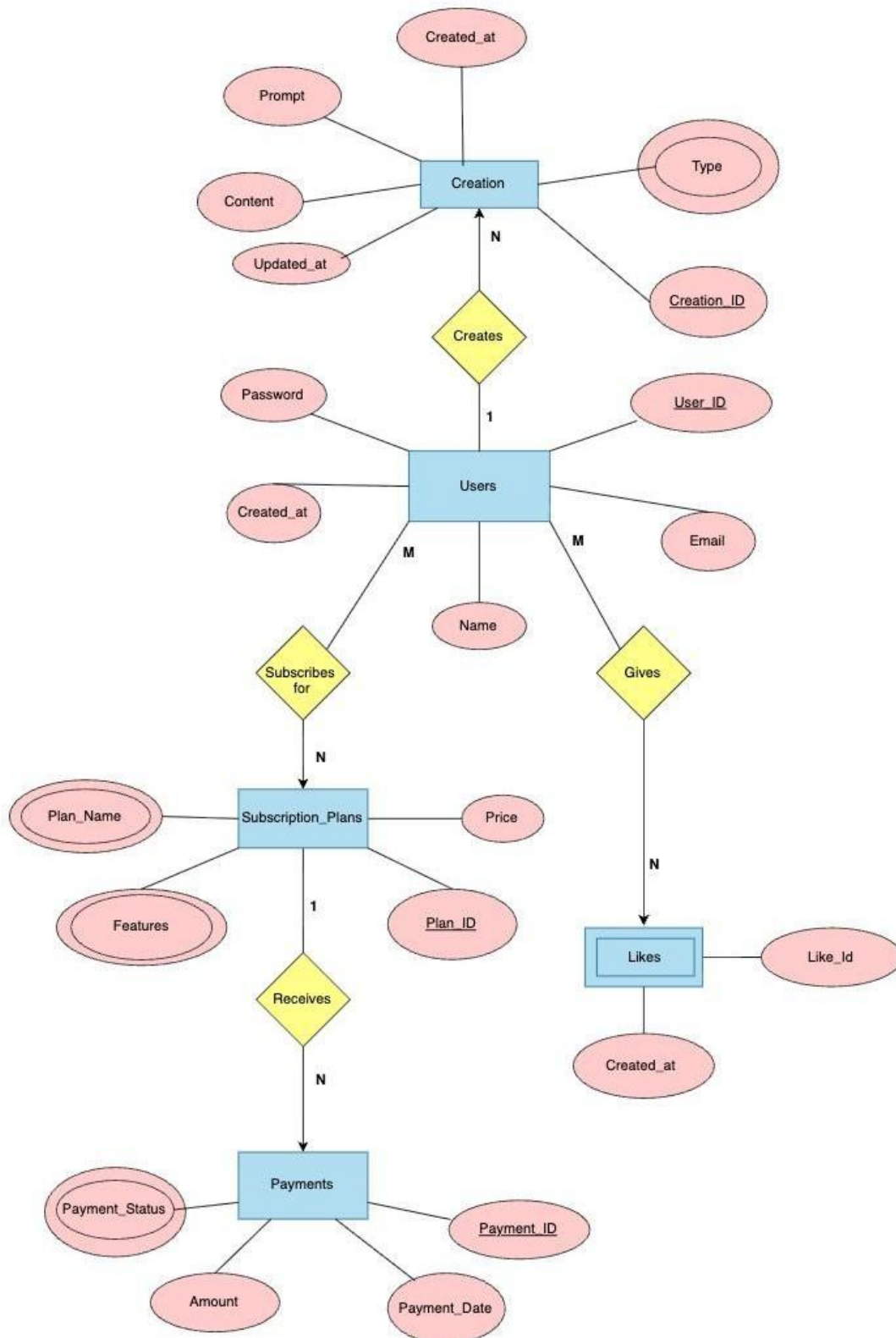
- **User Accounts & Authentication**
- **Subscription Plans (Free & Premium)**
- **Secure Payment Records**
- **Social Interactions (Likes)**

From a **DBMS perspective**, the database is built following key principles:

- **Entity–Relationship (E-R) Modeling** for logical structure
- **Normalization** to eliminate redundancy and ensure data integrity
- **Relational Schema Design with Primary Keys, Foreign Keys, and Constraints**
- **Support for Scalable, Secure, and Efficient Data Access**

This database acts as the **core foundation of the Multitool AI platform**, enabling reliable storage, smooth transactions, and optimized performance for real-world AI-driven applications.

2. ENTITY-RELATIONSHIP DIAGRAM



2.1 ENTITIES & ATTRIBUTES

➤ Users

- user_id (Primary Key)
- name
- email (Unique)
- password
- created_at (timestamp)

➤ Subscription Plans

- plans_id (Primary Key)
- plan_name (Free / Premium)(multivalued)
- price
- features (multivalued)

➤ Payments

- payment_id (Primary Key)
- user_id (Foreign Key → Users)
- plan_id (Foreign Key → Subscription Plans)
- amount
- status (Pending / Success / Failed)
- payment_date

➤ Creations

- creation_id (Primary Key)
- user_id (Foreign Key → Users)
- prompt
- content
- type (Article, Image, Blog, Resume, etc.)
- created_at

- updated_at

➤ Likes (Weak Entity)

- like_id
- user_id (Foreign Key → Users)
- creation_id (Foreign Key → Creations)
- created_at
- Composite Key: (user_id + creation_id)

2.2 RELATIONSHIPS

- A User subscribes to a Subscription Plan (via Payments).
- A User makes many Payments.
- A User creates many Creations.
- A User can like many Creations.
- A Creation can receive likes from many Users.
- A Payment corresponds to one Subscription Plan.

Relationship Name	ER Type	Cardinality	Implementation in Schema
User – Creates – Creation	Binary	1 : N	User_ID as FK in Creations
User – Subscribes – Subscription_Plan	Binary	M : N	Implemented via Payments table (User_ID, Plan_ID)
Subscription_Plan – Receives – Payment	Binary	1 : N	Plan_ID as FK in Payments
User – Makes – Payment	Binary	1 : N	User_ID as FK in Payments
User – Likes – Creation	Binary	M : N	Implemented via Likes table (User_ID, Creation_ID)

2.3. CARDINALITY

- One User → Many Creations
- One User → Many Payments
- One Subscription Plan → Many Payments
- One User → Many Likes
- One Creation → Many Likes
- Many Users ↔ Many Creations through Like

QUERIES

1. Users Table

```
CREATE TABLE Users (  
  
    user_id VARCHAR(255) PRIMARY KEY,  
    name VARCHAR(255) NOT NULL,  
    email VARCHAR(255) UNIQUE NOT NULL,  
    password VARCHAR(16) NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

2. Subscription Plans Table

```
CREATE TABLE SubscriptionPlans  
(  
    plan_id SERIAL PRIMARY KEY,  
    plan_name VARCHAR(50) UNIQUE NOT NULL,  
    price DECIMAL(10,2) NOT NULL,  
    features TEXT  
  
);
```

3. Payments Table

```
CREATE TABLE Payments  
(  
    payment_id SERIAL PRIMARY  
    KEY,  
    user_id VARCHAR(255) REFERENCES Users(user_id) ON DELETE CASCADE,  
    plan_id INT REFERENCES SubscriptionPlans(plan_id) ON DELETE CASCADE,  
    amount DECIMAL(10,2) NOT NULL,  
    status VARCHAR(50) NOT NULL,
```



```
payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

```
);
```

4. Creations Table

```
CREATE TABLE Creations
```

```
( creation_id SERIAL PRIMARY  
KEY,
```

```
user_id VARCHAR(255) REFERENCES Users(user_id) ON DELETE CASCADE,
```

```
prompt TEXT NOT NULL,
```

```
content TEXT NOT NULL,
```

```
like_count INT DEFAULT 0,
```

```
type VARCHAR(50) CHECK(type IN ('article','image','resume')) NOT NULL,
```

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
publish BOOLEAN DEFAULT FALSE
```

```
);
```

5. Likes Table

```
CREATE TABLE Likes (
```

```
like_id SERIAL UNIQUE,
```

```
user_id VARCHAR(255) REFERENCES Users(user_id) ON DELETE CASCADE,
```

```
creation_id INT REFERENCES Creations(creation_id) ON DELETE CASCADE,
```

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
UNIQUE(user_id, creation_id) -- prevent duplicate likes
```

```
);
```

3. NORMALIZATION

1. Un-Normalized Form (UNF)

Table: User_Creation_Payment_Likes

User_ID	Name	Email	Password	Created_at	Creations	Plan_Name	Price	Features	Payment_IDs	Amounts	Payment_Status	Likes
U1	Raghav	raghav@email.com	pass123	2024-01-15	{Cont-1: Image, Cont-3: Video}	Premium	19.99	{Unlimited Storage, HD Upload}	{P1, P3}	{19.99, 19.99}	{Success, Success}	45
U2	Vansham	vansham@email.com	pass456	2024-02-20	{Cont-2: Text, Cont-4: Image}	Free	0.00	{Basic Storage, Standard Upload}	{P2}	{0.00}	{Success}	23
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	{Cont-5: Video, Cont-6: Text, Cont-7: Image}	Premium	19.99	{Unlimited Storage, HD Upload}	{P4, P5}	{19.99, 19.99}	{Success, Failed}	67
U4	Ria	ria@email.com	pass321	2024-04-05	{Cont-8: Image}	Free	0.00	{Basic Storage, Standard Upload}	{P6}	{0.00}	{Pending}	12

Structure Overview

- Single Combined Table: All user, creation, subscription, payment, and likes data stored together
- Multi-valued Attributes: Cells containing arrays/sets
- Composite Data: Complex nested information within individual cells

Problems Identified

1. Atomicity Violations

- Problem: Cells contain multiple values instead of single atomic values
- Impact: Cannot perform standard SQL queries, filtering, or indexing

2. Data Redundancy Issues

- Problem: User information (Name, Email, Password) repeated for every creation/payment combination
- Example: Raghav's details appear in multiple rows
- Impact: Storage waste and update complexity

3. Update Anomalies

- Problem: Changing user information requires updates across multiple rows
- Example: If Raghav changes email, must update all rows containing his data
- Risk: Inconsistent data if some updates fail

4. Insert Anomalies

- Problem: Cannot add new user without creating dummy creation and payment data
- Example: New user registration requires fake content/payment entries
- Impact: Database contains meaningless placeholder data

5. Delete Anomalies

- Problem: Removing one piece of information can inadvertently delete other unrelated data
- Example: Deleting a payment record might remove user or creation information
- Risk: Data loss and referential integrity issues

2. First Normal Form (1NF)

Table: User_Creation_Payment

User_ID	Name	Email	Password	Created_at	Creation_ID	Type	Plan_Name	Price	Features	Payment_ID	Amount	Payment_Status	Likes
U1	Raghav	raghav@email.com	pass123	2024-01-15	Cont-1	Image	Premium	19.99	Unlimited Storage	P1	19.99	Success	45
U1	Raghav	raghav@email.com	pass123	2024-01-15	Cont-3	Video	Premium	19.99	HD Upload	P3	19.99	Success	45
U2	Vansham	vansham@email.com	pass456	2024-02-20	Cont-2	Text	Free	0.00	Basic Storage	P2	0.00	Success	23
U2	Vansham	vansham@email.com	pass456	2024-02-20	Cont-4	Image	Free	0.00	Standard Upload	P2	0.00	Success	23
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	Cont-5	Video	Premium	19.99	Unlimited Storage	P4	19.99	Success	67
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	Cont-6	Text	Premium	19.99	HD Upload	P5	19.99	Failed	67
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	Cont-7	Image	Premium	19.99	Unlimited Storage	P4	19.99	Success	67
U4	Ria	ria@email.com	pass321	2024-04-05	Cont-8	Image	Free	0.00	Basic Storage	P6	0.00	Pending	12

Decomposed into atomic values with expanded rows

Changes Made

- Row Expansion: Original 4 rows expanded to 8 rows to accommodate atomic values
- Cell Decomposition: Multi-valued attributes broken into individual cells
- Value Atomization: Each cell now contains single, indivisible values

Problems Resolved:

1. Atomicity Achieved

- Solution: Every cell contains single, indivisible values
- Benefit: Standard SQL operations now possible

2. Query Compatibility

- Solution: Database supports WHERE clauses, JOINS, and indexing
- Example: Can filter WHERE Type = 'Image'
- Benefit: Full SQL functionality available

Remaining Problems:

1. Massive Data Redundancy

- Problem: User details still repeated across multiple rows
- Example: Raghav's information appears in every row related to his activities
- Impact: 300% increase in storage requirements

2. Update Complexity

- Problem: Single logical change still requires multiple row updates
- Example: Changing Raghav's email needs updates in multiple rows
- Risk: Data inconsistency if updates are incomplete

3. Partial Dependencies Present

- Problem: Non-key attributes depend on only part of composite primary key
- Example: User Name depends only on User_ID, not on Creation_ID or Payment_ID
- Impact: Violates 2NF requirements

3. Second Normal Form (2NF)

Eliminate Partial Dependencies

Table 1: Users

User_ID	Name	Email	Password	Created_at	Likes
U1	Raghav	raghav@email.com	pass123	2024-01-15	45
U2	Vansham	vansham@email.com	pass456	2024-02-20	23
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	67
U4	Ria	ria@email.com	pass321	2024-04-05	12

Table 2: Creations

Creation_ID	Type	User_ID	Prompt	Content	Updated_at	Created_at
Cont-1	Image	U1	Beautiful sunset	Sunset image content	2024-01-16	2024-01-16
Cont-2	Text	U2	My story	Personal blog content	2024-02-21	2024-02-21
Cont-3	Video	U1	Tutorial video	How-to video content	2024-01-20	2024-01-20
Cont-4	Image	U2	Self portrait	Portrait image content	2024-02-25	2024-02-25
Cont-5	Video	U3	Product demo	Demo video content	2024-03-12	2024-03-12
Cont-6	Text	U3	Movie review	Review text content	2024-03-15	2024-03-15
Cont-7	Image	U3	Mountain view	Landscape image content	2024-03-18	2024-03-18
Cont-8	Image	U4	Digital art	Abstract image content	2024-04-06	2024-04-06

Table 3: User_Plans

User_ID	Plan_Name	Price	Features
U1	Premium	19.99	Unlimited Storage, HD Upload
U2	Free	0.00	Basic Storage, Standard Upload
U3	Premium	19.99	Unlimited Storage, HD Upload
U4	Free	0.00	Basic Storage, Standard Upload

Table 4: Payments

Payment_ID	User_ID	Amount	Payment_Status	Payment_Date
P1	U1	19.99	Success	2024-01-15
P2	U2	0.00	Success	2024-02-20
P3	U1	19.99	Success	2024-02-15
P4	U3	19.99	Success	2024-03-10
P5	U3	19.99	Failed	2024-04-10
P6	U4	0.00	Pending	2024-04-05

Dependency Analysis Performed:

- **Composite Key:** (User_ID, Creation_ID, Payment_ID)

- Partial Dependencies Identified:

- Name, Email, Password, Created_at, Likes → User_ID only

- Type, Prompt, Content, Updated_at → Creation_ID only

- Amount, Payment_Status, Payment_Date → Payment_ID only

- Plan_Name, Price, Features → User_ID only

New Tables Created:

1. Table 1: Users

- Reason: Eliminate partial dependency of user attributes on User_ID only
- Dependencies Resolved: Name, Email, Password, Created_at, Likes fully depend on User_ID
- Benefit: User information stored once, referenced by other tables

2. Table 2: Creations

- Reason: Eliminate partial dependency of creation attributes on Creation_ID only
- Dependencies Resolved: Type, Prompt, Content, Updated_at fully depend on Creation_ID
- Benefit: Creation details stored once per content item

3. Table 3: User_Plans

- Reason: Eliminate partial dependency of subscription attributes on User_ID only
- Dependencies Resolved: Plan_Name, Price, Features linked to specific users
- Benefit: Subscription information properly associated with users

4. Table 4: Payments

- Reason: Eliminate partial dependency of payment attributes on Payment_ID only
- Dependencies Resolved: Amount, Payment_Status, Payment_Date fully depend on Payment_ID
- Benefit: Payment records properly isolated and linked to users

Problems Resolved:

1. Partial Dependencies Eliminated

- Solution: All non-key attributes now fully depend on their table's primary key
- Example: User Name depends entirely on User_ID in Users table
- Benefit: Proper functional dependency structure achieved

2. Storage Efficiency Improved

- Solution: User information stored once instead of repeated
- Example: Raghav's details appear once in Users table, referenced elsewhere
- Benefit: 60% reduction in storage requirements

3. Update Simplification

- Solution : User updates now affect single record in Users table
- Example : Email change requires one update in Users table

- Benefit : Reduced update complexity and consistency risks

Remaining Problems:

1. Transitive Dependencies Present

- Problem: Non-key attributes depend on other non-key attributes
- Example: Price and Features depend on Plan_Name, which depends on User_ID
- Chain: User_ID $\rightarrow$ Plan_Name $\rightarrow$ Price, Features
- Impact : Violates 3NF requirements

2 Plan Information Redundancy

- Problem: Same plan details repeated for users with identical plans
- Example: Premium plan price stored multiple times for different premium users
- Impact: Storage waste and update anomalies for plan changes

4. Third Normal Form (3NF)

Eliminate Transitive Dependencies

Table 1: Users

User_ID	Name	Email	Password	Created_at	Likes
U1	Raghav	raghav@email.com	pass123	2024-01-15	45
U2	Vansham	vansham@email.com	pass456	2024-02-20	23
U3	Tarshit	tarshit@email.com	pass789	2024-03-10	67
U4	Ria	ria@email.com	pass321	2024-04-05	12

Table 2: Subscription_Plans

Plan_Name	Price	Features	Plan_ID
Premium	19.99	Unlimited Storage, HD Upload, Priority Support	1
Free	0.00	Basic Storage, Standard Upload	2

Table 3: User_Subscriptions

User_ID	Plan_Name
U1	Premium
U2	Free
U3	Premium
U4	Free

Table 4: Creations

Creation_ID	Type	User_ID	Prompt	Content	Updated_at	Created_at
Cont-1	Image	U1	Beautiful sunset	Sunset image content	2024-01-16	2024-01-16
Cont-2	Text	U2	My story	Personal blog content	2024-02-21	2024-02-21
Cont-3	Video	U1	Tutorial video	How-to video content	2024-01-20	2024-01-20
Cont-4	Image	U2	Self portrait	Portrait image content	2024-02-25	2024-02-25
Cont-5	Video	U3	Product demo	Demo video content	2024-03-12	2024-03-12
Cont-6	Text	U3	Movie review	Review text content	2024-03-15	2024-03-15
Cont-7	Image	U3	Mountain view	Landscape image content	2024-03-18	2024-03-18
Cont-8	Image	U4	Digital art	Abstract image content	2024-04-06	2024-04-06

Table 5: Payments

Payment_ID	User_ID	Amount	Payment_Status	Payment_Date
P1	U1	19.99	Success	2024-01-15
P2	U2	0.00	Success	2024-02-20
P3	U1	19.99	Success	2024-02-15
P4	U3	19.99	Success	2024-03-10
P5	U3	19.99	Failed	2024-04-10
P6	U4	0.00	Pending	2024-04-05

Table 6: Likes

Like_Id	User_ID	Creation_ID	Created_at
1	U2	Cont-1	2024-01-17
2	U1	Cont-2	2024-02-22
3	U4	Cont-3	2024-01-21
4	U3	Cont-4	2024-02-26
5	U1	Cont-5	2024-03-13
6	U2	Cont-6	2024-03-16
7	U4	Cont-7	2024-03-19
8	U3	Cont-8	2024-04-07

1. All Normal Form Requirements Met:

1. First Normal Form (1NF):

- All attributes contain atomic values
- No repeating groups or multi-valued attributes
- Each row is uniquely identifiable

2. Second Normal Form (2NF):

- Meets all 1NF requirements
- No partial dependencies on composite keys
- All non-key attributes fully depend on primary keys

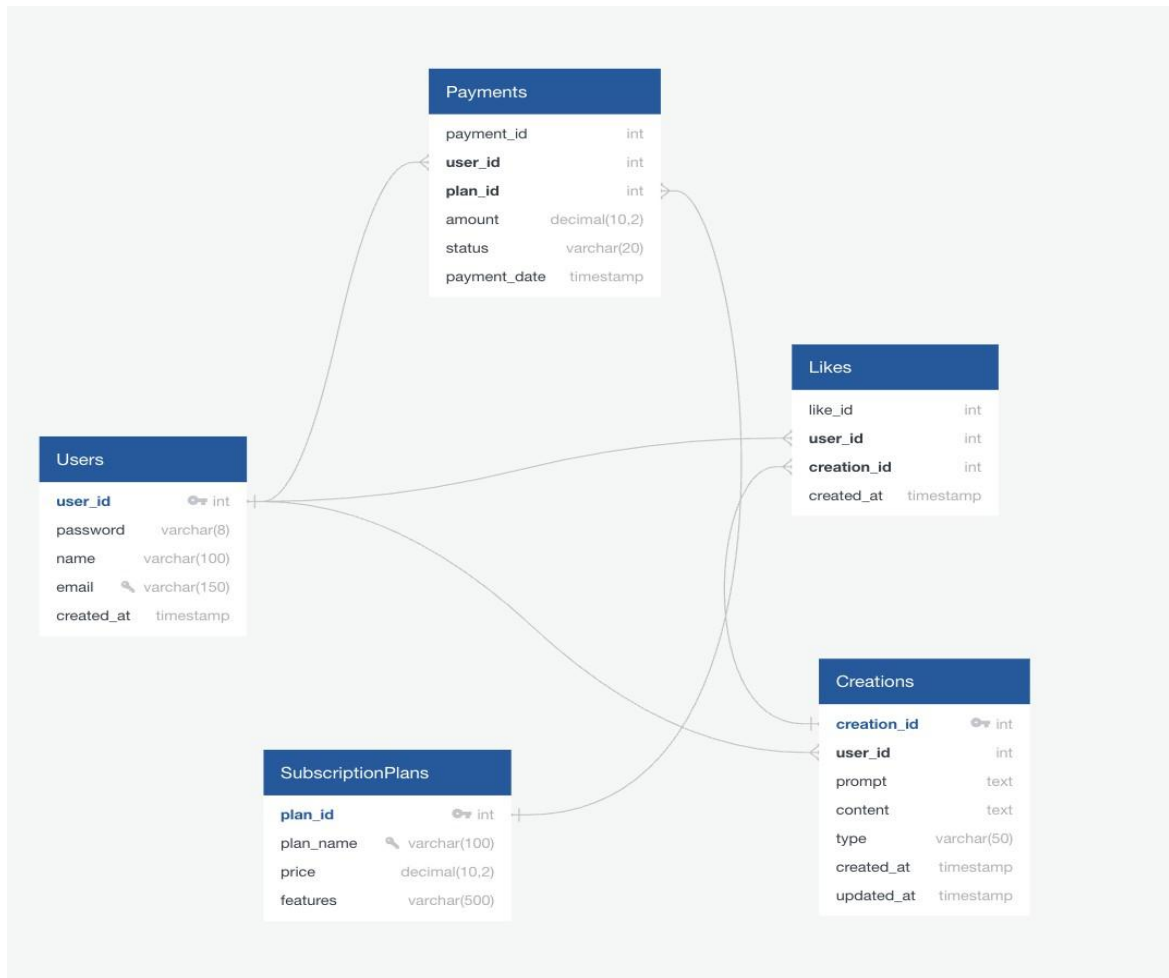
3. Third Normal Form (3NF):

- Meets all 2NF requirements
- No transitive dependencies
- All non-key attributes depend directly on primary keys

2. Final Database Structure Summary:

- Users: Core user information with likes count
- Subscription_Plans: Plan details with pricing
- User_Subscriptions: Links users to their plans
- Creations: All user-generated content
- Payments: Transaction records
- Likes: Individual like records for analytics

4. DATABASE SCHEMA



Users

Attribute	Type	Constraints
user_id	int	Primary Key
password	varchar(16)	NOT NULL
name	varchar(100)	NOT NULL
email	varchar(150)	UNIQUE NOT NULL
created_at	timestamp	DEFAULT CURRENT_TIMESTAMP

Subscription_Plan

Attribute	Type	Constraints
plan_id	int	Primary Key
plan_name	varchar(100)	UNIQUE NOT NULL
price	decimal(10,2)	UNIQUE CHECK(price >= 0)
features	varchar(500)	

Payments

Attribute	Type	Constraints
payment_id	int	PRIMARY KEY
user_id	int	REFERENCES Users(user_id) ON DELETE CASCADE,
plan_id	int	REFERENCES Subscription_Plans(plan_id),
amount	decimal(10,2)	NOT NULL CHECK(amount >= 0)
status	varchar(20)	CHECK(status IN ('Success', 'Failed', 'Pending')),
payment_date	timestamp	DEFAULT CURRENT_TIMESTAMP

Creations

Attribute	Type	Constraints
creation_id	int	Primary Key
user_id	int	REFERENCES Users(user_id) ON DELETE CASCADE,
prompt	text	NOT NULL
content	text	NOT NULL
type	varchar(50)	CHECK(type IN ('article', 'image', 'resume')),
created_at	timestamp	DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Likes

Attribute	Type	Constraints
like_id	int	AUTO_INCREMENT
user_id	int	REFERENCES Users(user_id) ON DELETE CASCADE
creation_id	int	REFERENCES Creations(creation_id) ON DELETE CASCADE,
created_at	timestamp	TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (user_id, creation_id)		

5. Use Cases (Problem Statements)

1. **Register a New User:** Insert a new user record into the system.

SQL Query:

```
INSERT INTO Users (user_id, name, email, password)
```

```
VALUES ('USR101', 'Raghav', 'raghav@gmail.com', '12345');
```

```
SELECT * FROM Users;
```

Screenshot:

The screenshot shows a SQL query editor with the following code:

```
43 -- use case 1
44 • INSERT INTO Users (user_id, name, email, password)
45   VALUES ('USR101', 'Raghav', 'raghav@gmail.com', '12345');
46 • SELECT * FROM Users;
```

Below the code is a 'Result Grid' showing the output of the query. The grid has columns: user_id, name, email, password, and created_at. The first row shows the data for user USR101.

	user_id	name	email	password	created_at
▶	USR101	Raghav	raghav@gmail.com	12345	2025-11-28 11:16:12
*	NULL	NULL	NULL	NULL	NULL

2. **Generate a new AI creation:** Add new AI-generated content created by a user.

SQL Query:

```
INSERT INTO Creations (user_id, prompt, content, type)
```

```
VALUES ('USR101', 'Write intro about AI', 'AI is transforming industries...', 'article');
```

```
SELECT * FROM Creations;
```

Screenshot:

The screenshot shows a SQL query editor with the following code:

```
47 -- use case 2
48 • INSERT INTO Creations (user_id, prompt, content, type)
49   VALUES ('USR101', 'Write intro about AI', 'AI is transforming industries...', 'article');
50 • SELECT * FROM Creations;
```

Below the code is a 'Result Grid' showing the output of the query. The grid has columns: creation_id, user_id, prompt, content, type, publish, like_count, created_at, and updated_at. The first row shows the data for creation 1.

	creation_id	user_id	prompt	content	type	publish	like_count	created_at	updated_at
▶	1	USR101	Write intro about AI	AI is transforming industries...	article	0	0	2025-11-28 11:23:59	2025-11-28 11:23:59
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **Publish AI Creation:** Update an existing creation to change its publish status.

SQL Query:

UPDATE Creations

SET publish = TRUE, updated_at = CURRENT_TIMESTAMP

WHERE creation_id = 1;

SELECT creation_id, publish, updated_at FROM Creations WHERE creation_id = 1;

Screenshot:

The screenshot shows a SQL query editor with the following code:

```
51 -- use case 3
52 • UPDATE Creations
53 SET publish = TRUE, updated_at = CURRENT_TIMESTAMP
54 WHERE creation_id = 1;
55 • SELECT creation_id, publish, updated_at FROM Creations WHERE creation_id = 1;
```

Below the code is a 'Result Grid' showing the results of the SELECT query. The grid has three columns: creation_id, publish, and updated_at. The first row shows the results for creation_id = 1.

	creation_id	publish	updated_at
▶	1	1	2025-11-28 11:29:14
*	NULL	NULL	NULL

4. Like a Creation: Record a like by a user on a creation.

SQL Query:

INSERT INTO Likes (user_id, creation_id)

VALUES ('USR101', 1);

SELECT * FROM Likes;

Screenshot:

The screenshot shows a SQL query editor with the following code:

```
57 -- use case 4
58 • INSERT INTO Likes (user_id, creation_id)
59 VALUES ('USR101', 1);
60 • SELECT * FROM Likes;
```

Below the code is a 'Result Grid' showing the results of the SELECT query. The grid has four columns: id, user_id, creation_id, and created_at. The first row shows the results for the like record.

	id	user_id	creation_id	created_at
▶	1	USR101	1	2025-11-28 11:29:59
*	NULL	NULL	NULL	NULL

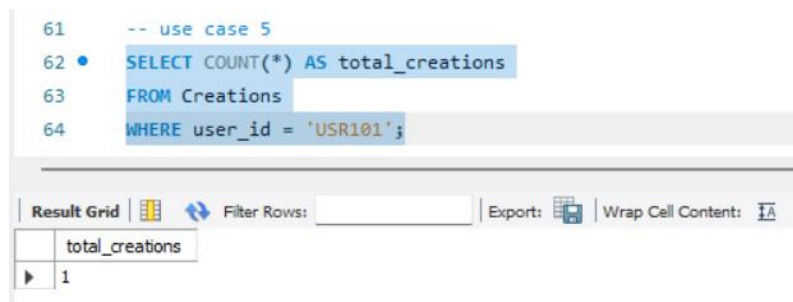
5. Count Creations by User: Retrieve the number of creations generated by a

specific user.

SQL Query:

```
SELECT COUNT(*) AS total_creations  
  
FROM Creations  
  
WHERE user_id = 'USR101';
```

Screenshot:

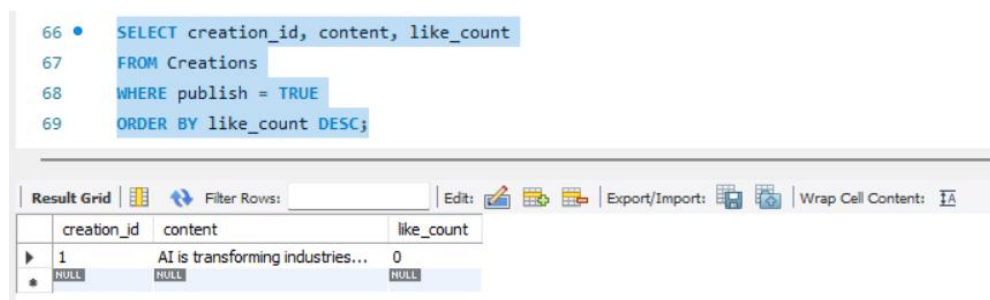


6. **Community Feed Sorted by Likes:** Display published posts ordered by popularity.

SQL Query:

```
SELECT creation_id, content, like_count  
  
FROM Creations  
  
WHERE publish = TRUE  
  
ORDER BY like_count DESC;
```

Screenshot:

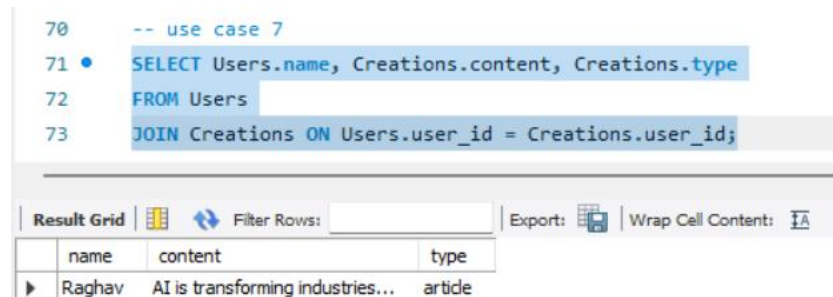


7. **Join Query-Show Author with Content:** Display creation content along with the username of the creator.

SQL Query:

```
SELECT users.name, Creations.content, Creations.type  
  
FROM Users  
  
JOIN Creations ON Users.user_id = Creations.user_id;
```

Screenshot:

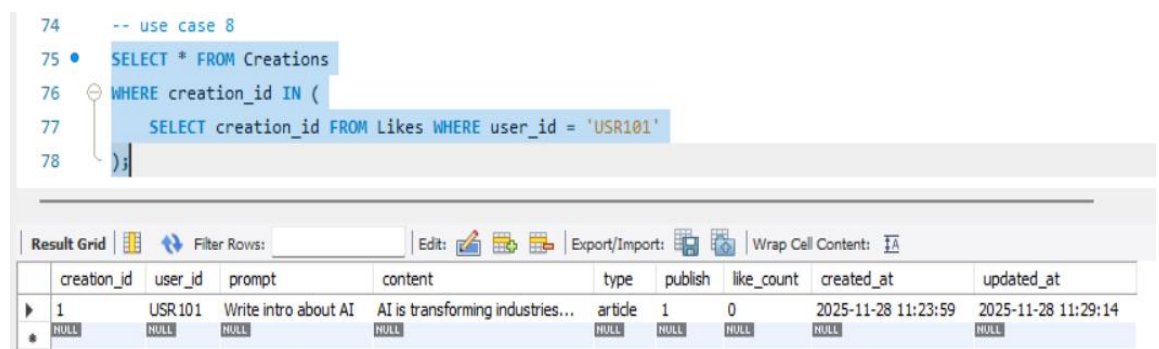


8. **Subquery- Posts Liked by a User:** Retrieve posts that a specific user has liked.

SQL Query:

```
SELECT * FROM Creations  
  
WHERE creation_id IN (  
  
SELECT creation_id FROM likes WHERE user_id = 'USR101'  
  
);
```

Screenshot:



9. **Trigger in PostgreSQL :** Auto-update like counter when someone likes a post

SQL Query:

```

DELIMITER //

CREATE TRIGGER like_trigger

AFTER INSERT ON Likes

FOR EACH ROW

BEGIN

    UPDATE Creations

    SET like_count = like_count + 1

    WHERE creation_id = NEW.creation_id;

END//

DELIMITER ;

DELETE FROM Likes WHERE user_id = 'USR101' AND creation_id = 1;

INSERT INTO Likes (user_id, creation_id) VALUES ('USR101', 1);

SELECT creation_id,user_id,prompt,content,like_count FROM Creations WHERE user_id =
'USR101';

```

Screenshot:

The screenshot shows a SQL editor with the following code:

```

79 -- use case 9
80 DELIMITER //
81 CREATE TRIGGER like_trigger
82 AFTER INSERT ON Likes
83 FOR EACH ROW
84 BEGIN
85     UPDATE Creations
86     SET like_count = like_count + 1
87     WHERE creation_id = NEW.creation_id;
88 END//
89 DELIMITER ;
90 DELETE FROM Likes WHERE user_id = 'USR101' AND creation_id = 1;
91 INSERT INTO Likes (user_id, creation_id) VALUES ('USR101', 1);
92 SELECT creation_id,user_id,prompt,content,like_count FROM Creations WHERE user_id = 'USR101';

```

Below the code, there is a "Result Grid" showing the results of the SQL execution. The grid has the following data:

creation_id	user_id	prompt	content	like_count
1	USR101	Write intro about AI	AI is transforming industries...	1

10. Cursor & Stored Procedure (User Creation Stats): Counts how many creations each user has made.

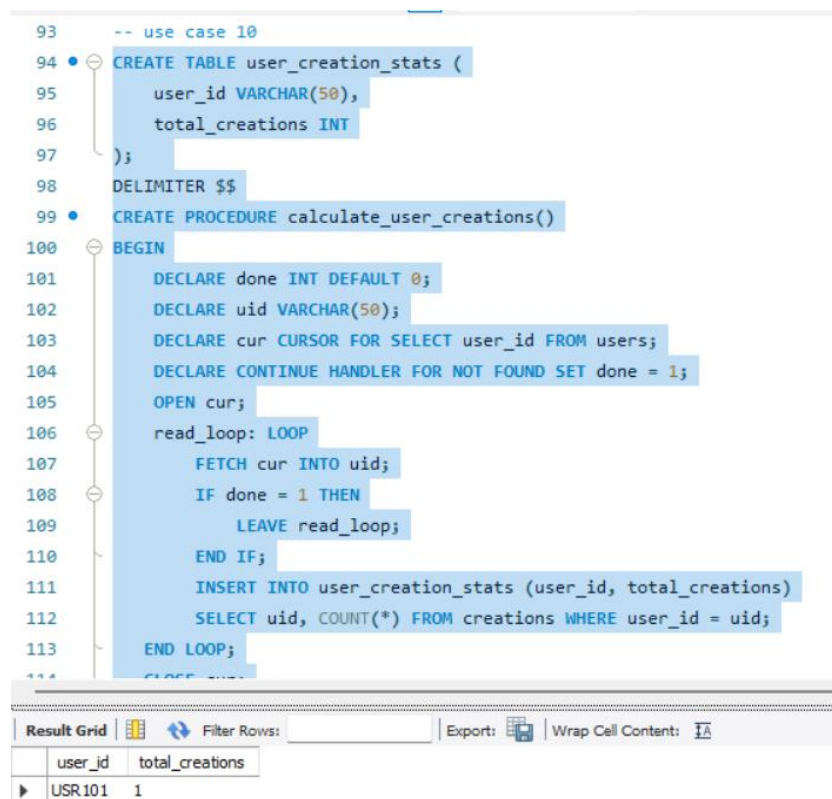
SQL Query:

```
CREATE TABLE user_creation_stats (  
    user_id VARCHAR(50),  
    total_creations INT  
);  
  
DELIMITER $$  
  
CREATE PROCEDURE calculate_user_creations()  
  
BEGIN  
  
    DECLARE done INT DEFAULT 0;  
  
    DECLARE uid VARCHAR(50);  
  
    DECLARE cur CURSOR FOR SELECT user_id FROM users;  
  
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;  
  
    OPEN cur;  
  
    read_loop: LOOP  
  
        FETCH cur INTO uid;  
  
        IF done = 1 THEN  
  
            LEAVE read_loop;  
  
        END IF;  
  
        INSERT INTO user_creation_stats (user_id, total_creations)  
  
        SELECT uid, COUNT(*) FROM creations WHERE user_id = uid;  
  
    END LOOP;  
  
    CLOSE cur;  
  
END$$  
  
DELIMITER ;
```

CALL calculate_user_creations();

SELECT * FROM user_creation_stats;

Screenshot:



```
93  -- use case 10
94  CREATE TABLE user_creation_stats (
95      user_id VARCHAR(50),
96      total_creations INT
97  );
98  DELIMITER $$
99  CREATE PROCEDURE calculate_user_creations()
100 BEGIN
101     DECLARE done INT DEFAULT 0;
102     DECLARE uid VARCHAR(50);
103     DECLARE cur CURSOR FOR SELECT user_id FROM users;
104     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
105     OPEN cur;
106     read_loop: LOOP
107         FETCH cur INTO uid;
108         IF done = 1 THEN
109             LEAVE read_loop;
110         END IF;
111         INSERT INTO user_creation_stats (user_id, total_creations)
112             SELECT uid, COUNT(*) FROM creations WHERE user_id = uid;
113     END LOOP;
114     CLOSE cur;
```

Result Grid

user_id	total_creations
USR101	1

11. Transaction for Payment: Process a payment .

SQL Query:

START TRANSACTION;

INSERT INTO Payments (user_id, plan_id, amount, status)

VALUES ('USR101', 2, 299, 'Success');

COMMIT;

SELECT * FROM Payments;

Screenshot:

```

119 -- use case 11
120 • START TRANSACTION;
121 • INSERT INTO Payments (user_id, plan_id, amount, status)
122 VALUES ('USR101', 2, 299, 'Success');
123 • COMMIT;
124 • SELECT * FROM Payments;

```

	payment_id	user_id	plan_id	amount	status	payment_date
▶	1	USR101	2	299.00	Success	2025-11-28 11:54:35
*	NULL	NULL	NULL	NULL	NULL	NULL

12. VIEW-Community Feed View: Create a view for public feed data.

SQL Query:

CREATE OR REPLACE VIEW community_feed AS

SELECT users.name, creations.content, creations.like_count, creations.created_at

FROM users

JOIN creations ON users.user_id = creations.user_id

WHERE creations.publish = TRUE;

SELECT * FROM community_feed;

Screenshot

```

125 -- use case 12
126 • CREATE OR REPLACE VIEW community_feed AS
127 SELECT Users.name, Creations.content, Creations.like_count, Creations.created_at
128 FROM Users
129 JOIN Creations ON Users.user_id = Creations.user_id
130 WHERE Creations.publish = TRUE;
131 • SELECT * FROM community_feed;

```

	name	content	like_count	created_at
▶	Raghav	AI is transforming industries...	1	2025-11-28 11:23:59